PROTOTYPE OFFER • HACKATHON TRACK: IMPACT • NOVEMBER 2026

# A VR training prototype for Unspoken Smiles programmes

Roadmap and proposed project: realistic 3D dental training simulations with interactive, to-spec equipment - built as a prototype to offer to the Unspoken Smiles Foundation.



Illustration, not a photograph and not a screenshot of the finished app. The prototype itself is described in the text.

> **Status, stated plainly:** No contact has been made with the Unspoken Smiles Foundation about this prototype, and no pilot, partnership or endorsement has been agreed. This document is an offer: try the prototype, tell us what is wrong, and decide whether a pilot is worth discussing. Nothing in it claims the Foundation built, approved or uses this work.

**What this is.** A separate VR training app, in the repository AGIFutureFoundation/UnspokenSmiled-VR, inspired by the Foundation's publicly described work - school-based oral health education, its Fellowship for sustainable dental careers, and preventive care in communities with limited access to a dentist. It is a demonstration prototype for a hackathon (NemulAI x Hack District, Impact track), not an official product.

**What it is not.** It does not licence or certify anyone, it does not train anyone on real patients, it collects no patient data, and it makes no clinical claims. Simulation here is practice only - a point the roadmap returns to in the licensure section, because the original request asked for a route to licence users and that route does not exist in VR.

## One working session first, the rest as an honest catalogue

The hackathon brief rewards one working feature over a long list of unfinished ones, so the prototype is scoped that way. Prevention and School Screening is built end to end as a playable session. The other worlds and stations are present as a browsable catalogue and wiki, clearly labelled as catalogue rather than playable, so a reviewer can see the full programme without being misled about what works today.

### What the prototype demonstrates now

- **One fully working session:** Prevention and School Screening - 9 steps, a coach guide, interactive props and checkpoints, up to 180 XP in the session scoring inherited from the earlier TradePath dental sessions.
- **Two further worlds in build:** Chairside Assisting Basics (Dental Operatory) and Sterilization and Operatory Setup (Sterilization Lab), each also structured as 9 steps.
- **The stack:** Meta's Immersive Web SDK (IWSDK) 1.0 - Three.js with a high-performance entity-component system, pre-built grab, ray and poke interaction for controllers or hands, locomotion, Havok physics, spatial audio and UIKitML spatial UI. Scaffolded with npm create @iwsdk@latest on Node 20.19 or later.
- **No-headset access:** the IWER desktop emulator and a magic-window fallback mean a judge, a teacher or a Foundation reviewer can open the same HTTPS address in a browser without a Quest headset. WebXR requires a secure context (HTTPS or localhost).
- **To-spec equipment:** the priority equipment in the spec table later in this document is modelled with its real operating ranges shown in-app, and with cycle timers and failure consequences rather than points alone.
- **Guides built in:** an in-app wiki covering how to enter each world, what each step teaches, controls, and demo limits; guide images; and a deep video walkthrough so a new user can follow the session unaided.
- **A coach dashboard in Demo mode,** labelled as sample data only - synthetic data, not real learners, with small groups suppressed.

> **Why IWSDK and not the existing code's own host layer?** The WebXR dental sims reviewed for this roadmap use raw WebXR with Three.js 0.160 and their own raycasting and locomotion, and their authors note they have not yet been exercised on Quest hardware. IWSDK replaces that host layer - grab, locomotion, physics and UI - while the sims' step data, scoring and station registry are ported across. One version check is required: IWSDK uses a super-three fork (0.181 in the review) against the sims' 0.160.

## The Foundation's work, as it describes it

The Foundation's About page states its mission as promoting both long-term oral health care and economic security in the world's most disadvantaged regions. Its founder, Jean Paul Laurent, was an NYU dentistry student who returned to Haiti after the 2010 earthquake to provide free dental care, and the organisation grew from that work. Its homepage frames the problem as nearly half the world's population suffering from oral disease, given there as 3.58 billion people, under the line "Thinking globally, acting locally", with a current focus on Costa Rica and an ambition across the hemisphere.

### Programmes relevant to a training prototype

- **Zero Cavity Initiative:** vetted professionals and Fellows go into schools to provide free screenings, education and preventive care. The disease figures on that page are the CDC's, not the Foundation's own measurements.
- **Fellowship:** the 2026 cohort is described as 30 fellows aged 17-38 from Costa Rica, Panama, Peru and Nicaragua, in its third cohort, with more than 1,000 schoolchildren screened across 8 public schools in San Jose Central since inception, and ULACIT faculty supervising weekly at the new clinic with students earning academic credit. An earlier published model recruited women aged 18-35 for five week-long courses plus a two-month externship, supervised by licensed professionals, each fellow paired with a dentist and a hygienist mentor.
- **Preventive and Green Dental Clinic, Paseo Colon, San Jose:** described by the Foundation as the world's first fully preventive and green dental clinic - that is their claim, not independently verified here. It was set to open in September 2025, and a Facebook post announced a grand opening on 8 September 2025. No post-opening status page was found on the site.
- **A June 2025 screening** is reported as 403 children across 6 preschools, 61.3 percent enrolled, with 190 children (47.1 percent) having untreated cavities - the clearest concrete picture of the screening work the prototype rehearses.



Illustration of the school-screening setting the first working session rehearses: screening, brushing education and a mobile outreach base.

## Numbers that conflict - and stay conflicting in this document

Published figures for the Foundation do not agree with each other, so this roadmap quotes them with their source rather than merging them into one settled number. The founding year is given as 2014 in most sources, with a pilot in May 2014 in Jacmel, Haiti, but as March 2011 in L'Union Suite; the Foundation's own About page gives no year. Children served appear as more than 10,000 in 9 countries on the homepage, more than 9,000 in 9 countries on the About page, and more than 10,000 in 10 countries with 52,000 hygiene products in a June 2025 release. Fellows in 2025 appear as 21 in that release and as 40 in a November 2025 Facebook post. The homepage counters rendered as zero in the text review, so no counter value is cited here at all. Any pilot discussion should start by asking the Foundation which figures it wants used.

Partners the Foundation names publicly include ULACIT, IADS, FIGS, Axiom2, the ADA and Henry Schein. Its social presence is founder-led and celebratory - Instagram for the organisation and for the founder, Facebook, and LinkedIn - and its volunteer page states that it does not accept unsolicited internship or volunteer inquiries, naming fundraising as the route it offers. Nothing reviewed on its site or social channels mentions this prototype, its builders, or any VR training. That absence is why this document is framed as an offer, not a collaboration.

# What already exists in code - and what does not

The starting repository is empty. AGIFutureFoundation/UnspokenSmiled-VR was created on the day this work began, with a single commit containing a two-line README ("Training simulator developed for UnspokenSmiles Programs"), an MIT licence file and an Android .gitignore. There is no code, engine, scene or dental content to port. The prototype is a first build, and the value in the wider codebase is in content and patterns, not in a ready-made app.

## Two complete dental programmes: 37 unique stations

The strongest source is the vr-safety-training project, which carries two dental packs in its WebXR branch. The Dental Careers pack (19 stations) covers four-handed dentistry, radiography including a full-mouth series, the sterilisation technician cycle, the lab bench, orthodontic, oral surgery, implant and endodontic assisting, front-office coordination, infection control audit, school screening outreach, teledentistry, digital scanning and CAD/CAM, trauma response, and coding and billing. The Dental Hygiene pack (19 stations) covers operatory turnover, instrument reprocessing, sharps exposure, radiograph safety, periodontal charting, ultrasonic scaling, aerosol management, fluoride and sealants, nitrous oxide monitoring, chairside emergency, mobile outreach, the pediatric visit and oral cancer screening. Only one station - patient intake screening - appears in both, giving 37 unique stations. A third pack, a dental library of 36 stations, carries no certification field. Sampled simulations in these packs have 14 to 15 graded steps plus 2 fault steps each, declared as data rather than hard-coded, and rendered from procedural primitives - there are no hosted 3D model files and no equipment specification numbers in them. That last gap is exactly what the spec work in this roadmap fills.

## The authored clinic world, games and collectibles

SmartCiti.X contains an authored smiles world: a registry of 37 stations across 15 zones and 9 rooms - reception (5 stations), operatory (7), hygiene (7), imaging (3), sterilisation (5), surgery (5), lab (1), careers (1) and outreach (3) - plus 6 games and 16 collectibles. The games are directly reusable as prevention teaching: brushing for 120 seconds across 4 quadrants, flossing, a plaque hunt over 9 spots, snack sorting, a drinks ranking, and handwashing for 20 seconds. The 16 tooth-fairy collectibles each reveal a public-health point. The registry carries the disclaimer "General guidance, not medical or dental advice", which the prototype keeps. One discrepancy is flagged rather than smoothed over: the world generator's own description says 8 collectibles while the registry counts 16; the wiki uses 16 and notes the mismatch. The generator also hard-codes a local reference-clone path, so it is not portable as-is, and its build check is fail-closed on exactly 37 stations.

## Patterns worth copying, from three codebases

- **SmartCiti.X skill graph and sequencer:** 3,663 skills arranged as 11 strands by 3 tiers, sequenced with weights favouring the zone of proximal development (0.45), then review (0.25), leverage (0.15), modality (0.10) and novelty (0.05), targeting a challenge band of 0.70-0.85, with mastery gates at 0.95 plus 3 unaided successes and a 6-rung hint ladder in which a hinted pass is never scored as unaided. The trades registry contains no dental skills, so the prototype authors a new, small dental registry in the same shape - with safety and infection-control fundamentals gating everything else.
- **Cognition.X curriculum shape:** a Flow Zone library of 17,450 blocks, where one block is one doable skill plus a transfer check on real material; and a Simulation Studio pattern of a brief plus 5 decision points with 3 options each, a consequence and a debrief. Its rule is adopted verbatim: simulation is practice, never a check, never a credential.
- **WorldForge world and quest shape:** a world described as data, with districts - here Reception, Operatory, Sterilisation and Screening Hall - an indoor lighting preset, 5-step quests (talk, collect, visit, talk, report back), coach characters, and collectibles. Its local engine is flat-screen third-person Three.js with no WebXR, so the VR layer comes from IWSDK, not from WorldForge. Its separate dental robot pack is telemetry - about 13,000 messages - not a learner simulation, and is not used as one.

> **Licence caution.** Neither SmartCiti.X nor vr-safety-training carries a root licence for its own code (only third-party components are licensed, under CC0 or MIT). The safe course, followed here, is to link to that work and port the step-data pattern and station structure, not to copy code wholesale. Two unrelated external repositories named "unspokensmiles" were also checked: they are a Turkish Django donation app, not connected to the Foundation or to this work, and are not used.

## The proposed project: three worlds, one catalogue, one wiki

The app is a single IWSDK project with a clinic hub. From the hub a learner enters a world, is met by a coach character, works through graded steps with grabbable equipment, and returns to report back - the 5-step quest loop from WorldForge, with each step scored deterministically. An answer-checking or coaching layer may comment, but it never changes the score: in the source simulations, scoring is deterministic (+100 for a correctly identified hazard, -25 for a false positive) and that separation is preserved, because a training record that a language model can alter is not a training record.

| World | Setting | Core skills rehearsed | Status in prototype |
|---|---|---|---|
| Prevention and School Screening | Outreach Hub: school hall, screening bay, brushing station, mobile van lot | Set up a screening bay; run the brushing and flossing games; sort snacks and drinks; record a screening result; handwash timing | Fully working session |
| Chairside Assisting Basics | Dental Operatory | Four-handed instrument transfer; suction placement; tray set-up; patient positioning; operatory turnover | In build |
| Sterilization and Operatory Setup | Sterilization Lab | Instrument reprocessing chain; ultrasonic cleaning; packaging; autoclave cycle and indicators; sharps handling; logbook | In build |
| Catalogue and wiki | Clinic district linking all rooms | All 37 stations browsable with their room, programme and step outline; 6 games; 16 collectibles | Browsable, labelled catalogue |

### How a session runs

1. **Brief.** The coach states the task, the safety point and what "good" looks like, in plain language and, following the authored world's example, with localisation in mind - the source world ships in 8 locales.
2. **Lesson and checklist.** A short lesson, then a visible checklist the learner works against, following the station pattern of lesson, checklist, tasks and quiz, then Mark Complete.
3. **Graded steps with faults.** Each simulation carries graded steps plus deliberately planted fault steps - a pouch sealed incorrectly, a sharps container overfilled, a screening form left unsigned - which the learner must catch. Missing a fault has a stated consequence in the debrief, not just a lost point.
4. **Debrief and log.** The session writes a learner progress event - who, which station, which steps, unaided or hinted, faults caught. The earlier station yard lost progress on reload, so persistence is an explicit build requirement.

### Wiki, images and video as part of the product

Guides are built alongside the app: a wiki page per world and station, guide images for controls and each room, and one deep video walkthrough of the working session. Public programme text comes from the Foundation's public pages; sponsor Firecrawl is the named route for turning those pages into clean structured text, and CleanShot for annotated capture and the walkthrough video.

## Equipment built to spec, so the practice transfers

A simulation teaches whatever its objects do. If the autoclave accepts any temperature and any order of steps, it teaches that sterilisation is approximate. Every priority asset is therefore built as an IWSDK entity with grabbable or interactable components, its real operating range displayed in-app, a named reference model, and a consequence when it is used outside that range. Ranges vary by manufacturer, so the app shows the range and labels its reference model rather than presenting one brand as the specification.



Illustration of the priority asset set. In the app each item is a separate interactive entity with its own controls and limits.

| Asset | Specification to model | Teaching point |
|---|---|---|
| Dental chair and delivery unit | Chair height about 430-720 mm; delivery up to 4 handpiece positions; air 0.5-0.8 MPa, water 0.2-0.3 MPa | Positioning, four-handed working, checking pressures before starting |
| Operating light | LED 8,000-30,000 lux, 4,000-5,500 K, colour rendering index 85-90 or better | Aiming without dazzling; cure-safe mode |
| Class B autoclave | EN 13060 pre-vacuum; 132-134 degrees C for 3-10 minutes; drying 15-20 minutes; handpieces never above 135 degrees C | The highest-value simulation. Chemical indicator every load; spore test at least weekly and for every implantable load |
| Ultrasonic cleaner | 35-48 kHz, 2-6 litre tank, 5-10 minutes | Cleaning is not sterilisation - the distinction is scored |
| Suction and syringe | High-volume evacuator about 3.3 SCFM; saliva ejector; 3-way syringe | Placement that keeps the field clear without obstructing |
| Ultrasonic scaler and handpieces | Scaler 28 kHz plus or minus 3, 3-20 W, water coolant | Light touch, coolant always on, no dry scaling |

## Equipment to spec, continued

| Asset | Specification to model | Teaching point |
|---|---|---|
| Intraoral X-ray set | 60-70 kV, 2-8 mA; receptor, apron and positioning kit | Simulated only, never emitting. ALARA, placement, angulation and retake logic |
| Packaging, cassette and logbook | Pouches, cassettes, indicator and spore-test kit, sterilisation log | Traceability: an unlogged load is an unfinished load |
| PPE, handwash and waste | Handwash 20 seconds; sharps and amalgam waste streams | Infection control as habit, not as a quiz answer |
| Mobile van and screening kit | Portable chair, light, screening forms, fluoride and education materials | The outreach setting the Foundation's screening work actually uses |
| Lab bench and CAD/CAM scanner | Specification not verified in this research - modelled generically and flagged "to be confirmed" | No invented numbers. This asset waits for a clinician or manufacturer source |

Realism also means order matters. In the sterilisation chain, loading the autoclave before ultrasonic cleaning, or skipping the indicator, fails the step with the real-world consequence explained in the debrief. Timers run at real duration in assessed mode, with an accelerated option clearly labelled as practice mode.

# Licensure-ready, not licencing

> **This app cannot licence users, and the roadmap does not pretend otherwise.** Licensure and certification are granted by a state board, a national examination body or an accredited school, after an accredited programme, supervised clinical hours on real patients, and in-person written and clinical examinations. No XP total, badge, rank or completed VR run in this prototype is a credential, a completion record for any external body, or evidence of clinical competence.

What the prototype can legitimately do is prepare a learner for that external path: rehearse the sequences an examiner or supervisor will expect, build the vocabulary, and map practice to the domains the real examinations cover, with the external step named on the page where the learner meets it.

## The external pathways, as researched

- **Dental assistant, United States - DANB Certified Dental Assistant:** three examinations - General Chairside, Radiation Health and Safety, and Infection Control - reached either through a CODA-accredited programme or through at least 3,500 hours of work experience.
- **California Registered Dental Assistant:** a state licence requiring an approved programme of at least 800 hours, or 15 months with at least 1,280 hours of experience, plus Radiation Safety, Coronal Polishing, Sealants and Infection Control courses, then the RDA examination.
- **Dental hygienist, United States:** a CODA-accredited programme, then the National Board Dental Hygiene Examination (350 questions, pass at 75 or above), then a regional or state clinical examination, then a state board licence.
- **Costa Rica:** no verified primary source was found in this research for a standalone dental assistant or hygienist licence pathway. This roadmap does not present one. It needs to be checked directly with the Colegio de Cirujanos Dentistas de Costa Rica and with ULACIT before any Costa Rica pathway is described to learners.

## How the app supports the path without crossing the line

- Every station page ends with a "where this leads" panel naming the external programme or examination domain it prepares for, and stating that VR practice does not count as clinical hours.
- Mastery gates use the SmartCiti pattern - a high threshold plus repeated unaided successes - and are labelled "ready to be supervised on this task", never "qualified".
- The source packs' own "certification state" is a software unlock rule inside that repository, not a licence, and the prototype renames it to avoid any confusion.
- The guide character is presented as a demo guide, not a clinician, following the safety wording already used in the related Diagnostic.X work: screening and teaching support only.

# What a pilot would give the Foundation - and what it would need

## What the Foundation would get from a pilot

- **Fellowship training support:** fellows could rehearse operatory set-up, four-handed assisting and the sterilisation chain repeatedly, before they are in a clinic, without consuming instruments, pouches or supervisor time on first attempts.
- **School-screening rehearsal:** the working session mirrors the screening-and-education visit - set up, screen, record, teach brushing - so a fellow's first real school visit is not their first attempt at the sequence.
- **Sterilisation and operatory practice on demand:** the highest-risk routine in a small clinic, available for deliberate practice with faults planted on purpose, which a real clinic cannot safely do.
- **A coach dashboard in Demo mode:** a view of which stations a cohort has practised, where hints were needed and which faults were missed - in the pilot's first phase using sample or anonymised practice data agreed in advance, with small groups suppressed.
- **The wiki and walkthrough video,** which double as onboarding material for new fellows and as an explanation the Foundation can share with its own partners.

## What we need from the Foundation to proceed

1. **Clinical review:** a practising dental assistant or registered dental hygienist to review the step sequences and the equipment specifications and say what is wrong - the prototype's authors are not clinicians, and the sequences should not be taught until a clinician has checked them.
2. **Faculty review, if they choose:** ULACIT faculty review, since ULACIT already supervises fellows at the clinic; this is an invitation, not an assumption.
3. **A small pilot cohort:** a handful of fellows or trainees, with a named supervisor, a defined number of sessions, and agreed success measures - completion, faults caught, supervisor judgement - rather than open-ended use.
4. **A written agreement before any use** of the Foundation's name or logo, its learners, or any patient-facing material. Until that exists, the prototype carries its own demo labelling and does not use the Foundation's branding.

## What is explicitly not offered

- No licence, certificate or credential of any kind, and no claim that practice in the app counts toward one.
- No clinical training on real patients, and no patient contact of any kind through the app.
- No partnership, endorsement, affiliation or joint project - none exists, and none is implied by this document.
- No patient data: the app neither collects, stores nor needs it, and the dashboard works on practice events only.
- No medical or dental advice to the public. In-app guidance is training guidance, labelled as general guidance.

## Roadmap: seven phases from spec bible to licensure bridge

| Phase | Build | Done when |
|---|---|---|
| 0 - Spec bible | Freeze the equipment specification table, the dental skill registry in SmartCiti shape, and the station-to-skill map for all 37 stations. Flag unverified items rather than filling them in. | A clinician reviewer can mark the bible up; every number has a source or a "to be confirmed" flag |
| 1 - Operatory | Clinic hub, Reception and Operatory districts, chair, delivery, light and suction as IWSDK entities; four-handed assisting session; desktop fallback. | A judge completes the session in a browser with no headset |
| 2 - Sterilisation chain | Ultrasonic cleaner, packaging, Class B autoclave with real cycle logic, indicators, spore-test workflow and logbook; fault steps planted. | Wrong order fails with a consequence debrief; a correct chain logs completely |
| 3 - Scaling and hygiene | Scaler and handpieces to spec, periodontal charting, fluoride and sealants, aerosol management. | Hygiene pack stations playable with the same gate rules |
| 4 - Radiography, simulated | X-ray head, receptor and apron; placement, angulation and retake logic; ALARA teaching throughout. Nothing emits. | A learner can explain why a retake is or is not justified |
| 5 - Lab, outreach and screening | Lab bench (once its specification is verified), mobile van, school-screening kit; the Prevention and School Screening session completed and filmed for the deep walkthrough. | The hackathon demo: one session, start to debrief, on video |
| 6 - Licensure bridge | Exam-domain map to DANB General Chairside, Radiation Health and Safety and Infection Control, and to NBDHE domains; external links to DANB, CODA and state boards; "VR practice is not clinical hours" on every page. | A learner can see, from any station, the real external step it prepares for |

## Risks and open questions, kept visible

- **Source-code licences:** the two most useful codebases have no root licence for their own code. Port patterns and data structures, link rather than copy, and get permission before anything more.
- **Hardware:** the source WebXR sims were not exercised on Quest hardware by their authors. IWSDK's emulator covers development, but a Quest 3 session must be tested before any live demonstration is promised.
- **Conflicting public numbers** for the Foundation (founding year, children served, fellow counts) must be resolved with the Foundation, not by picking the most flattering figure.
- **Unverified specifications:** lab bench and scanner specifications, and the Costa Rica licensure pathway, are open items. They stay flagged until a primary source is checked.
- **Count mismatches in source data:** 8 versus 16 collectibles, and a dental spotlight world described in one codebase against 37 stations in another, are recorded as mismatches in the wiki rather than silently harmonised.
- **Clinical accuracy is the gating risk.** Without a practising clinician's review, this remains a hackathon prototype, however polished it looks.

## Sources the reader can check

Unspoken Smiles Foundation - website and About, Zero Cavity Initiative and Fellowship pages, unspokensmiles.org; its public Facebook, Instagram and LinkedIn posts as cited in the review. Code: AGIFutureFoundation/UnspokenSmiled-VR (empty starter repository); the vr-safety-training project's dental careers and dental hygiene packs and WebXR branch; SmartCiti.X smiles registry, authored stations and world generator; Cognition.X Flow Zone and Simulation Studio; WorldForge worlds data. Technology: Meta Immersive Web SDK documentation, developers.meta.com/vr/develop/iwsdk, and iwsdk.dev. Equipment and licensure: EN 13060 for Class B sterilisers; ISO 7494 for dental units; DANB certification pathways; California Dental Board RDA requirements; NBDHE and CODA programme information. Specific figures in this document are attributed in the text to the page or pack they came from, with conflicts shown rather than resolved.

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End of roadmap. Prototype offer only - no contact has been made with the Unspoken Smiles Foundation, and no pilot, partnership or endorsement has been agreed. Demo only; not affiliated; general guidance, not medical or dental advice.